



WishPool

Birthday Greetings Application

Product Requirements Document (PRD)

Version 1.0 | February 2026

Stack: FastAPI · Nuxt 3 · Supabase · Tailwind CSS

1. Executive Summary

WishPool is a web application that allows users to create personalised birthday greeting pages for celebrants. Friends and family are invited to contribute heartfelt messages and photos; the creator then shares a public, login-free link with the birthday person so they can enjoy all the wishes in one beautiful place.

Attribute	Detail
App Name	WishPool
Version	1.0 (MVP)
Document	Product Requirements Document (PRD)
Date	February 2026
Frontend	Nuxt 3 (Vue 3 · SSR/CSR hybrid)

Backend	FastAPI (Python 3.11+)
Database	Supabase (PostgreSQL)
Storage	Supabase Storage
Auth	JWT + Google OAuth 2.0
UI Framework	Tailwind CSS 3

2. Goals & Success Metrics

2.1 Primary Goals

- Allow authenticated users to create and manage birthday greeting pages.
- Enable friends and family to contribute personalised messages to a page.
- Deliver a beautiful, public, zero-login birthday reveal link to the celebrant.
- Support Google Sign-In as a frictionless authentication option.
- Provide fast, SEO-friendly public pages via server-side rendering (SSR).

2.2 Success Metrics (MVP)

Metric	Target	Priority
Page creation success rate	> 95%	High
Message submission success rate	> 98%	High
Public page load time (LCP)	< 2.5 s	High
Mobile responsiveness	100%	High
Google OAuth success rate	> 99%	Medium
Image upload success rate	> 97%	Medium
Social share preview accuracy	100%	Medium

3. User Personas

3.1 The Organiser (Page Creator)

Organiser

Role: Signs up, creates the greeting page, invites contributors, shares the reveal link.

Goals: Create a memorable surprise for the birthday person.

Pain Points: Hard to collect messages from scattered friends; wants one tidy place.

Tech Level: Comfortable with social apps; may be non-technical.

Actions: Sign up / log in → Create page → Upload photos → Share contributor link → Share reveal link.

3.2 The Contributor

Contributor

Role: Receives the contributor link, signs up or logs in, posts a message.

Goals: Write a heartfelt birthday message quickly and easily.

Pain Points: Does not want to download an app; wants a simple form.

Tech Level: Any level — must be very easy to use.

Actions: Click link → Sign up / log in → Write message → Submit.

3.3 The Celebrant (Birthday Person)

Celebrant

Role: Receives the public reveal link and views all birthday messages.

Goals: Enjoy a surprise page filled with wishes from loved ones.

Pain Points: Forced sign-up would ruin the surprise and experience.

Tech Level: Any level — zero friction required.

Actions: Click reveal link → View page with photos & messages (no login needed).

4. Functional Requirements

4.1 Authentication & Authorisation

Sign Up

- User can register with email and password.

- Password must be at least 8 characters, including one number and one special character.
- Email must be verified before full access is granted.
- Duplicate email addresses are rejected with a clear error message.

Log In

- User can log in with email and password.
- Failed login shows a generic error (do not reveal whether email exists).
- Session persists via JWT stored in httpOnly cookies.
- Access token expires after 30 minutes; refresh token expires after 7 days.

Google OAuth 2.0

- User can sign up or log in with a Google account ("Sign in with Google" button).
- On first Google login, a user record is created automatically.
- Avatar URL and display name are pulled from Google profile.
- Google account can be linked to an existing email/password account.

Log Out

- User can log out from any authenticated page.
- Logout invalidates the refresh token server-side.

Authorisation Rules

Resource	Action	Who Can Perform
Greeting Page	Create	Any authenticated user
Greeting Page	Read (full)	Creator only
Greeting Page	Edit / Delete	Creator only
Greeting Page	View (public)	Anyone (no login)
Message	Post	Any authenticated user
Message	Edit / Delete	Author only
Images	Upload	Creator only
Contributor Link	Access	Any authenticated user

4.2 Greeting Page Management

Create Greeting Page

- Required fields: Celebrant full name, date of birth.
- Optional fields: Cover photo (single image), additional gallery images (up to 10).
- System auto-generates two unique tokens on creation:

- `share_token` — used in the contributor link (`/greetings/{token}`)
- `public_token` — used in the celebrant reveal link (`/view/{token}`)
- Both tokens are cryptographically random (UUID v4 or 32-char hex).
- Creator is redirected to the page dashboard after creation.

Edit Greeting Page

- Creator can update celebrant name, birth date, and images at any time.
- Images can be added or removed individually.
- Changes are reflected immediately on the contributor and public pages.

Delete Greeting Page

- Creator can permanently delete a page.
- Deletion cascades to all associated messages and images.
- A confirmation dialog is shown before deletion.

Dashboard

- Creator sees a list of all their greeting pages.
- Each card shows: celebrant name, birth date, message count, and creation date.
- Quick-copy buttons for both the contributor link and the reveal link.
- Empty state shown when no pages exist, with a call-to-action to create one.

4.3 Image Uploads

- Accepted formats: JPEG, PNG, WebP.
- Maximum file size: 5 MB per image.
- Maximum images per page: 10 (including cover photo).
- Images are stored in Supabase Storage under a per-page folder.
- Uploaded images are publicly accessible via CDN URL.
- Invalid format or oversized files show a clear error message.
- Upload progress indicator is shown during upload.

4.4 Message / Greeting System

Post a Message (Contributor Page)

- User must be authenticated to post a message.
- Each user can post exactly one message per greeting page.
- Message text is required; minimum 1 character, maximum 1000 characters.
- Character count is shown in real time.
- Submitted messages appear immediately on the contributor page.

Edit / Delete Own Message

- A contributor can edit their message at any time before the reveal link is shared.
- A contributor can delete their own message.

Message Display

- Each message card shows: contributor's avatar, display name, message text, and relative timestamp (e.g. '2 hours ago').
- Messages are ordered chronologically (oldest first on the contributor page).
- On the public reveal page, messages are displayed in a visually rich grid or card layout.

4.5 Link Sharing

Contributor Link

- Format: `https://wishpool.app/greetings/{share_token}`
- Visiting this link prompts login/signup if the user is not authenticated.
- After authentication, user is redirected back to the contributor page.
- The page shows the celebrant name, photos, and the message submission form.

Reveal Link (Public)

- Format: `https://wishpool.app/view/{public_token}`
- No login required — fully public.
- Displays all messages, celebrant name, photos, and birth date.
- Page is server-side rendered (SSR) for SEO and social sharing.
- Open Graph meta tags (`og:title`, `og:description`, `og:image`) are populated from page data.

5. Pages & Routes

Route	Page Name	Render	Auth
/	Homepage / Landing	SSR	Public
/login	Login	CSR	Public
/signup	Sign Up	CSR	Public
/dashboard	My Greeting Pages	CSR	Auth
/dashboard/create	Create Page	CSR	Auth
/edit/[id]	Edit Page	CSR	Auth
/greetings/[token]	Contributor Page	SSR	Auth

/view/[token]	Public Reveal Page	SSR	Public
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5.1 Page Descriptions

Homepage (/)

- Marketing / landing page explaining the app.
- Sections: Hero, How It Works (3 steps), Features, Call to Action.
- SSR for SEO — Google should index and rank this page.
- CTA buttons: "Get Started Free" (→ /signup) and "Sign In" (→ /login).

Login (/login) & Sign Up (/signup)

- Centered card layout, no navbar.
- Email/password form + "Sign in with Google" button.
- Guest middleware: already-authenticated users are redirected to /dashboard.

Dashboard (/dashboard)

- List of all greeting pages created by the logged-in user.
- Each card: celebrant name, birth date, message count, copy-link buttons.
- Floating "+" button or top-right CTA to create a new page.

Create Page (/dashboard/create)

- Multi-step form or single scrollable form.
- Fields: Celebrant Name, Birth Date, Cover Image, Additional Images.
- Image upload with drag-and-drop support.
- Submit creates the page and redirects to dashboard.

Edit Page (/edit/[id])

- Pre-populated form with existing page data.
- Image management: view existing, upload new, delete individual images.
- Save changes button with loading state.

Contributor Page (/greetings/[token])

- Shows celebrant name, photos (carousel or grid), and existing messages.
- Authenticated users see a message form.
- Unauthenticated users see a prompt to sign in.
- Users who already posted see their message with edit/delete options.
- SSR for initial HTML (the share link may be posted on social media).

Public Reveal Page (/view/[token])

- No login required.
- Full celebrant name, birth date, photo gallery.

- All messages displayed in a beautiful card layout.
- SSR with Open Graph meta tags for rich social media previews.
- Confetti or celebration animation on first load (client-side only).

6. API Requirements

6.1 Authentication Endpoints

Method & Path	Description	Auth Required
POST /api/v1/auth/signup	Register with email & password	No
POST /api/v1/auth/login	Login, receive JWT	No
POST /api/v1/auth/google	Google OAuth callback	No
POST /api/v1/auth/logout	Invalidate refresh token	Yes
POST /api/v1/auth/refresh	Refresh access token	Yes
GET /api/v1/auth/me	Get current user profile	Yes

6.2 Greeting Page Endpoints

Method & Path	Description	Auth Required
GET /api/v1/greeting-pages	List pages for current user	Yes
POST /api/v1/greeting-pages	Create a new greeting page	Yes
GET /api/v1/greeting-pages/{id}	Get page details (owner only)	Yes
PUT /api/v1/greeting-pages/{id}	Update page details	Yes (owner)
DELETE /api/v1/greeting-pages/{id}	Delete page + cascade	Yes (owner)
GET /api/v1/greeting-pages/share/{share_token}	Get page by contributor token	Yes
GET /api/v1/greeting-pages/public/{public_token}	Get public page data	No

6.3 Message Endpoints

Method & Path	Description	Auth Required
GET /api/v1/messages/{page_id}	Get all messages for a page	No (public) / Yes (private)
POST /api/v1/messages	Post a new message	Yes
PUT /api/v1/messages/{id}	Edit own message	Yes (author)
DELETE /api/v1/messages/{id}	Delete own message	Yes (author)

6.4 Upload Endpoints

Method & Path	Description	Auth Required
POST /api/v1/uploads/image	Upload image to Supabase Storage	Yes (page owner)
DELETE /api/v1/uploads/image/{id}	Delete image from storage	Yes (page owner)

6.5 Standard API Response Format

Success Response

```
{ "success": true, "data": { ... }, "message": "Operation successful" }
```

Error Response

```
{ "success": false, "error": { "code": "NOT_FOUND", "message": "Greeting page not found", "details": {} } }
```

7. Database Schema

7.1 users

Column	Type	Constraints	Description
id	UUID	PK, default uuid_generate_v4()	Unique user ID
email	VARCHAR(255)	UNIQUE, NOT NULL	Login email
password_hash	VARCHAR(255)	NULLABLE	Null for OAuth users

full_name	VARCHAR(255)	NULLABLE	Display name
google_id	VARCHAR(255)	UNIQUE, NULLABLE	Google OAuth subject ID
avatar_url	TEXT	NULLABLE	Profile picture URL
created_at	TIMESTAMP	DEFAULT NOW()	Account creation date
updated_at	TIMESTAMP	DEFAULT NOW()	Last profile update

7.2 greeting_pages

Column	Type	Constraints	Description
id	UUID	PK	Unique page ID
creator_id	UUID	FK → users(id)	Page owner
celebrant_name	VARCHAR(255)	NOT NULL	Birthday person name
celebrant_birthdate	DATE	NOT NULL	Birth date
share_token	VARCHAR(100)	UNIQUE, NOT NULL	Contributor link token
public_token	VARCHAR(100)	UNIQUE, NOT NULL	Reveal link token
cover_image_url	TEXT	NULLABLE	Cover photo URL
created_at	TIMESTAMP	DEFAULT NOW()	Page creation date
updated_at	TIMESTAMP	DEFAULT NOW()	Last update date

7.3 celebrant_images

Column	Type	Constraints	Description
id	UUID	PK	Unique image ID
greeting_page_id	UUID	FK → greeting_pages(id)	Parent page
image_url	TEXT	NOT NULL	Supabase Storage URL
display_order	INT	DEFAULT 0	Sort order for gallery
created_at	TIMESTAMP	DEFAULT NOW()	Upload date

7.4 messages

Column	Type	Constraints	Description
id	UUID	PK	Unique message ID
greeting_page_id	UUID	FK → greeting_pages(id)	Parent page
user_id	UUID	FK → users(id)	Message author
message_text	TEXT	NOT NULL, max 1000 chars	Message content
created_at	TIMESTAMP	DEFAULT NOW()	Post date
updated_at	TIMESTAMP	DEFAULT NOW()	Last edit date
UNIQUE		greeting_page_id + user_id	One message per user per page

8. Non-Functional Requirements

8.1 Performance

- Public pages (SSR) must achieve a Largest Contentful Paint (LCP) < 2.5 seconds.
- API responses must return within 500 ms under normal load.
- Image CDN delivery via Supabase Storage for fast global access.
- Lazy loading for images on the gallery and message list.

8.2 Security

- All API endpoints use HTTPS in production.
- Passwords are hashed with bcrypt (cost factor ≥ 12).
- JWT tokens are stored in httpOnly, Secure, SameSite=Strict cookies.
- CORS restricted to the frontend origin only.
- Rate limiting on auth endpoints: 5 requests per minute per IP.
- File uploads validated for MIME type and file size server-side.
- SQL injection prevention via Supabase client parameterised queries.
- XSS prevention via Vue template auto-escaping (no raw v-html on user content).

8.3 Accessibility

- WCAG 2.1 Level AA compliance.
- All interactive elements are keyboard-navigable.
- Colour contrast ratio ≥ 4.5:1 for normal text.

- All images have descriptive alt attributes.
- Forms include proper labels and ARIA attributes.
- Error messages are announced to screen readers.

8.4 Responsiveness

- Mobile-first design — fully functional on screens 320px and wider.
- Breakpoints: sm (640px), md (768px), lg (1024px), xl (1280px).
- Touch-friendly tap targets (min 44×44px).
- Images and galleries adapt to screen size.

8.5 SEO

- Homepage is server-side rendered with full meta tags.
- Contributor page (/greetings/[token]) is SSR with basic Open Graph tags.
- Public reveal page (/view/[token]) has full Open Graph tags:
 - og:title — "Happy Birthday {Celebrant Name}!"
 - og:description — "{N} friends sent you birthday wishes!"
 - og:image — Celebrant cover photo
 - og:type — website
- robots.txt and sitemap.xml included.

8.6 Scalability

- Stateless FastAPI backend — horizontally scalable.
- Supabase handles database connection pooling.
- Static assets and images served via CDN.
- No server-side sessions — JWT-based auth.

9. UI / UX Requirements

9.1 Design System

Attribute	Value
Primary Font	Inter (Google Fonts)
Code Font	JetBrains Mono
Primary Color	#7C3AED — Purple 700
Secondary Color	#DB2777 — Pink 600

Accent Color	#0EA5E9 — Sky 500
Success	#16A34A — Green 600
Warning	#D97706 — Amber 600
Danger / Error	#DC2626 — Red 600
Background	#F9FAFB — Gray 50
Border Radius	8px (cards), 6px (inputs), 9999px (pills)
Transition Speed	200ms ease-in-out

9.2 Component Library (Reusable UI)

- UIButton — variants: primary, secondary, outline, ghost; sizes: sm, md, lg; loading state.
- UiInput — label, placeholder, error message, disabled state, required indicator.
- UiCard — clickable and non-clickable variants, elevated shadow.
- UiModal — title, body slot, footer slot, close on backdrop click.
- UiAlert — info, success, warning, error variants.
- UiSpinner — inline and full-page loading variants.
- UiToast — auto-dismiss notifications (success, error, info).

9.3 User Feedback States

- Loading — spinner or skeleton loader for all async operations.
- Success — green toast notification with checkmark.
- Error — red toast or inline error message with clear description.
- Empty — illustrated empty state with guidance text and a CTA.
- Confirmation — modal dialog for destructive actions (delete page, delete message).

9.4 Animations

- Page transitions: fade + slight upward slide (200ms).
- Card hover: subtle shadow lift and border colour change.
- Modal: fade + scale in (300ms).
- Public reveal page: confetti burst on first load (client-side only).
- Toast notifications: slide in from top-right corner.

10. Technology Stack

10.1 Frontend

Package	Version	Purpose
nuxt	^3.10	SSR/CSR hybrid Vue framework
vue	^3.4	UI component framework
pinia	^2.1	State management
tailwindcss	^3.4	Utility-first CSS
@vueuse/core	^10.9	Vue composable utilities
@nuxtjs/google-fonts	^3	Font loading
vue-sonner	^1.0	Toast notifications

10.2 Backend

Package	Version	Purpose
fastapi	^0.110	REST API framework
uvicorn[standard]	^0.29	ASGI server
supabase	^2.4	Database & storage client
pydantic	^2.6	Data validation & schemas
pydantic-settings	^2.2	Config from env vars
python-jose[cryptography]	^3.3	JWT token handling
passlib[bcrypt]	^1.7	Password hashing
python-multipart	^0.0.9	File upload support
authlib	^1.3	Google OAuth 2.0
httpx	^0.27	Async HTTP client

10.3 Infrastructure

Service	Purpose
Supabase	PostgreSQL database, Auth (optional), Storage (images)
Vercel / Netlify	Nuxt 3 frontend hosting (SSR support)
Render / Railway	FastAPI backend hosting
GitHub Actions	CI/CD pipeline (lint, test, deploy)

Cloudflare CDN	Static asset delivery (via Supabase Storage)
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11. Rendering Strategy (Nuxt 3)

Why Hybrid Rendering?

SSR pages are rendered on the server, delivering full HTML to the browser immediately.

This is critical for the public reveal page (social sharing previews) and the homepage (SEO).

CSR pages are rendered in the browser — better for interactive, authenticated dashboards.

Route	Mode	Reason	Notes
/	SSR	SEO, marketing landing page	Open Graph tags for app sharing
/login	CSR	No SEO needed, interactive form	Guest middleware applied
/signup	CSR	No SEO needed, interactive form	Guest middleware applied
/dashboard	CSR	Behind auth, real-time data	Auth middleware applied
/dashboard/create	CSR	Behind auth, form-heavy	Auth middleware applied
/edit/[id]	CSR	Behind auth, form-heavy	Owner check middleware
/greetings/[token]	SSR	Shareable link, social preview	Auth redirect on message submit
/view/[token]	SSR	Public, full OG tags, celebrant UX	No auth required

12. Environment Variables

12.1 Backend (.env)

Variable	Example Value	Description
SUPABASE_URL	https://xxx.supabase.co	Supabase project URL
SUPABASE_ANON_KEY	eyJhbGc...	Supabase anon/public key

SUPABASE_SERVICE_ROLE_KEY	eyJhbGc...	Supabase service role key
SECRET_KEY	32-char random string	JWT signing secret
ALGORITHM	HS256	JWT algorithm
ACCESS_TOKEN_EXPIRE_MINUTES	30	Access token TTL
REFRESH_TOKEN_EXPIRE_DAYS	7	Refresh token TTL
GOOGLE_CLIENT_ID	xxx.apps.googleusercontent.com	Google OAuth client ID
GOOGLE_CLIENT_SECRET	GOCSPX-...	Google OAuth client secret
FRONTEND_URL	http://localhost:3000	Allowed CORS origin
ENVIRONMENT	development	"development" or "production"

12.2 Frontend (.env)

Variable	Example Value	Description
NUXT_PUBLIC_API_URL	http://localhost:8000/api/v1	Backend API base URL
NUXT_PUBLIC_GOOGLE_CLIENT_ID	xxx.apps.googleusercontent.com	Google OAuth client ID
NUXT_PUBLIC_APP_URL	http://localhost:3000	Frontend base URL

13. Development Phases

Phase 1 — Foundation (Week 1–2)

- Backend: FastAPI project setup, Supabase connection, database schema.
- Backend: User model, Pydantic schemas, JWT auth endpoints.
- Frontend: Nuxt 3 setup, Tailwind CSS, folder structure.
- Frontend: Auth store (Pinia), login and signup pages.
- Integration: Connect frontend auth to backend auth API.

Phase 2 — Core Features (Week 3–4)

- Backend: Greeting page CRUD endpoints, token generation.
- Backend: Image upload to Supabase Storage.

- Backend: Message CRUD endpoints, one-per-user constraint.
- Frontend: Dashboard page, create/edit page forms.
- Frontend: Image uploader component with preview.
- Frontend: Contributor page with message form.

Phase 3 — Public & Polish (Week 5–6)

- Frontend: Public reveal page with SSR and Open Graph tags.
- Frontend: Confetti animation, celebration UI.
- Frontend: Link sharing UI (copy to clipboard, share buttons).
- Backend: Google OAuth 2.0 integration.
- Frontend: Google Sign-In button.

Phase 4 — Quality & Deploy (Week 7–8)

- Error handling, loading states, empty states across all pages.
- Mobile responsiveness audit and fixes.
- Accessibility audit (keyboard, screen reader, contrast).
- Backend unit tests (pytest).
- Frontend component tests (Vitest).
- CI/CD pipeline setup (GitHub Actions).
- Production deployment (Vercel + Render or Railway).

14. Future Enhancements (Post-MVP)

Feature	Description
GIF / sticker support	Allow contributors to attach GIFs or stickers to messages.
Video messages	Contributors record or upload short video greetings.
Email notifications	Notify creator when new messages are posted.
Page themes	Choose from multiple visual themes for the reveal page.
Scheduled reveal	Set a date/time for the reveal link to go live.
Download as PDF	Creator can download the greeting page as a PDF keepsake.
Anonymous messages	Option to allow contributors to post without logging in.
Multiple celebrants	Create a page for multiple people (e.g. couple birthday).
Admin dashboard	Super-admin view for moderation and analytics.
Analytics	Creator sees how many people viewed the reveal page.

15. Glossary

Term	Definition
Organiser	Authenticated user who creates and manages a greeting page.
Contributor	Authenticated user who posts a birthday message on a page.
Celebrant	The birthday person who receives the reveal link.
Greeting Page	The core entity: a page dedicated to one birthday celebrant.
share_token	URL-safe random token used in the contributor link.
public_token	URL-safe random token used in the public reveal link.
SSR	Server-Side Rendering — HTML generated on the server per request.
CSR	Client-Side Rendering — HTML generated in the browser via JavaScript.
JWT	JSON Web Token — a signed, self-contained auth token.
OAuth	Open standard for delegated authentication (e.g. Google Sign-In).
Supabase	Open-source Firebase alternative providing PostgreSQL and Storage.
Pinia	Official state management library for Vue 3.
OG Tags	Open Graph meta tags that control social media link previews.
MVP	Minimum Viable Product — the first releasable version.
PRD	Product Requirements Document — this document.